

CoM3D-ACE: Reciprocal Evidence-Graph Tracklet Refinement for Aerial Multi-Object Tracking

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Abstract

Aerial multi-object trackers often split one identity into several local tracklets when small targets are occluded or weakly resolved. We present CoM3D-ACE as a deterministic post-tracking refinement layer. It builds a directed candidate graph from temporally disjoint, same-class tracklets, applies fixed image-plane and appearance gates, retains only reciprocal predecessor-successor proposals, and rejects component merges that would create two observations with one identity in the same frame. No detector box is added, removed, or interpolated. We evaluate frozen ByteTrack, OC-SORT, and Deep OC-SORT outputs on all 50 sequences of the official MMOT test release. With common oracle AABBs, refinement raises equal-sequence-mean IDF1 by 2.28, 2.29, and 2.18 points and reduces total identity switches by 2,545, 1,678, and 585, respectively. With detections from the official MMOT YOLO11L-3ch checkpoint, the corresponding IDF1 gains are 1.65, 1.62, and 1.00 points, with 1,389, 1,059, and 324 fewer identity switches. On the 38 test sequences not used in our earlier analysis, paired sequence-bootstrap intervals for all six IDF1 gains remain above zero. Strong controlled alternatives nevertheless achieve higher mean accuracy: geometry-only greedy linking leads for ByteTrack, while partial Hungarian assignment leads for OC-SORT and Deep OC-SORT. Moreover, the fixed policy degrades IDF1 on held-out M3OT streams, where all admitted links with known identity are false. The evidence therefore supports a reproducible MMOT post-tracking improvement over no refinement, but not universal transfer or superiority over global assignment.

Keywords: aerial multi-object tracking, tracklet refinement, evidence graph, reciprocal association, small aerial objects

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1. Introduction

Aerial video provides wide-area coverage for traffic monitoring, inspection, and emergency response, but its targets are frequently small in native image coordinates. In the complete MMOT test release used here, 53.4% of annotated boxes have area below 32^2 pixels and 43.1% have a short side below 16 pixels. At this scale, a short occlusion, a missed detection, or a weak crop can interrupt a trajectory even when the target reappears only a few frames later. The resulting identity fragmentation affects association metrics and downstream trajectory analysis without necessarily changing localization.

Modern online trackers attack this problem within the frame-to-frame update. ByteTrack recovers low-score detections during association [1]; OC-SORT improves motion reasoning through observation-centric updates [2]; and Deep OC-SORT supplements that design with appearance cues [3]. A distinct option is to retain these tracker outputs and link compatible tracklets after tracking. Post-processing is attractive because it can be applied to archived results and compared under exactly the same detections and local trajectories. It is also risky: one false link can merge two otherwise valid identities.

This work asks a deliberately narrow question: can a fixed evidence-graph rule improve temporal identity continuity without changing the upstream detector or tracker? We instantiate each local tracklet as an EvidenceToken, form directed candidate relations between temporally ordered fragments, and retain only mutual predecessor–successor choices that pass fixed geometry and appearance gates. A component-level frame-overlap test prevents an accepted chain from assigning the same identity to two boxes in one frame. Fragments with no admissible relation remain unchanged. The method uses no camera pose, 3D reconstruction, ground-truth identity, learned link classifier, or box interpolation.

Figure 1 summarizes the evaluated path. It also makes the scope explicit: CoM3D-ACE begins after a local tracker has produced tracklets and ends with relabeled temporal trajectories. It does not claim global cross-UAV identity or autonomous view planning in this study.

We make three contributions:

- We specify a tracker-agnostic, deterministic tracklet-refinement rule with fixed temporal, image-plane, appearance, reciprocal, and component constraints, together with explicit unchanged output for unsupported links.

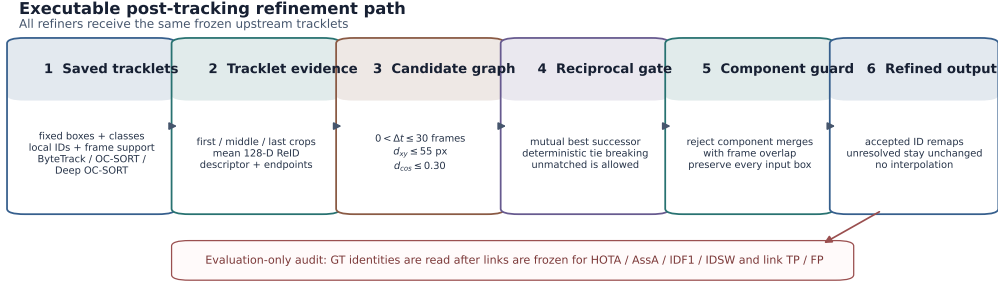


Figure 1: Evaluated CoM3D-ACE temporal pipeline. Frozen local tracklets are encoded, gated into a directed candidate graph, filtered by reciprocal predecessor–successor selection, and merged only when component frame supports remain disjoint. Unmatched tracklets and every box, class, and confidence value are preserved. The schematic was produced by a deterministic Matplotlib script prepared with OpenAI Codex assistance and verified against the implementation by the authors.

- We evaluate three frozen tracker families under both common oracle AABBs and one shared official detector output on all 50 MMOT test sequences. A frozen 38-sequence confirmation partition separates new evidence from the 12 sequences analyzed previously.
- We report direct greedy, partial-Hungarian, and official AFLink controls; component ablations; link error versus coverage; runtime; and an adverse held-out M3OT transfer result. These analyses define where the fixed rule helps and where stronger assignment or domain-specific validation is required.

The contribution is thus an auditable post-tracking refinement study, not a claim that a deterministic graph replaces state-of-the-art online or global tracking systems.

2. Related Work

2.1. Aerial tracking and evaluation

Aerial MOT combines the usual identity-association problem with small object size, camera motion, and crowded views. MMOT provides multispectral UAV video with oriented annotations and dense, small targets [4]; M3OT adds multi-drone RGB and infrared sequences [5]. We evaluate individual temporal streams from these releases. This isolates within-stream tracklet refinement and must not be interpreted as global cross-camera identity evaluation.

ByteTrack associates high- and low-confidence detections [1]. OC-SORT emphasizes observation-centric motion updates under occlusion [2], and Deep OC-SORT introduces adaptive appearance association [3]. We use all three as frozen upstream trackers rather than modifying their update rules. HOTA decomposes tracking accuracy into detection and association components [6]; IDF1 evaluates identity continuity through global trajectory matching [7]. Both are reported with CLEAR identity switches.

2.2. Post-tracking tracklet linking

Global association can repair trajectories that an online tracker has split. GIAOTracker combines local aerial tracking with a global linking stage [8]. StrongSORT++ includes AFLink, a learned post-linker, and Gaussian-smoothed interpolation as distinct modules [9]. Our comparison invokes the released AFLink model and checkpoint only; interpolation is excluded because every method must preserve the common box set.

Graph-based MOT methods learn structured relations among detections or tracklets. Neural message passing can optimize global associations across a detection graph [10], while city-scale multi-camera systems combine appearance and spatiotemporal constraints [11, 12]. CoM3D-ACE is narrower: it is a deterministic post-processing rule with a fixed pretrained descriptor and no training on the target benchmark.

2.3. Selective and constrained association

Every post-linker trades recovered continuity against false merges. Selective prediction motivates leaving uncertain cases unresolved rather than forcing a decision [13], but a heuristic distance threshold is not automatically a calibrated risk estimate [14, 15]. We therefore report accepted-link error together with coverage at the fixed operating point and across a descriptive sweep. The component frame-overlap rule is a hard output-validity constraint; it is not a probabilistic confidence guarantee.

3. Method

3.1. Frozen tracklets as evidence tokens

Let an upstream tracker produce a set of local tracklets $\mathcal{T} = \{T_i\}_{i=1}^K$. Tracklet T_i contains observations $z_{it} = (t, b_{it}, s_{it}, c_i)$, where t is the released frame index, $b_{it} = (x, y, w, h)$ is an axis-aligned box in native image pixels, s_{it} is the upstream confidence, and c_i is the class. In the evaluated adapter,

classes are processed independently, so every candidate relation is same-class by construction.

An EvidenceToken for tracklet i stores T_i , its frame support $F_i = \{t : z_{it} \in T_i\}$, first and last endpoint boxes, and an optional appearance descriptor $\mathbf{f}_i \in \mathbb{R}^{128}$. The descriptor is computed from at most three direct tracker-box crops: the first, middle, and last observations. Each crop is resized to 128×256 , transformed by the fixed BaseReID evaluation normalization, and L2-normalized after encoding. We average the available crop features and normalize the result again. No ground-truth matching is used to select or encode a crop.

3.2. Directed candidate graph

A directed edge $i \rightarrow j$ is temporally eligible only if the two frame supports are disjoint, T_i ends before T_j begins, and

$$1 \leq g_{ij} = \min F_j - \max F_i \leq G, \quad G = 30. \quad (1)$$

Let $\mathbf{p}_i^{\text{end}}$ and $\mathbf{p}_j^{\text{start}}$ be the centers of the endpoint boxes. The image-plane and appearance distances are

$$d_{ij}^{xy} = \|\mathbf{p}_i^{\text{end}} - \mathbf{p}_j^{\text{start}}\|_2, \quad d_{ij}^{app} = 1 - \mathbf{f}_i^\top \mathbf{f}_j. \quad (2)$$

The fixed candidate gates are $d_{ij}^{xy} \leq 55$ pixels and $d_{ij}^{app} \leq 0.30$. These values were fixed from M3OT development data before the held-out tests used here. They are not adapted to sequence name, modality, tracker, or MMOT identity labels.

3.3. Reciprocal proposal and component constraint

For every source tracklet, the method chooses the outgoing candidate with the lexicographically smallest tuple

$$q_{ij} = (d_{ij}^{xy}, d_{ij}^{app}, g_{ij}, i, j). \quad (3)$$

It independently chooses the smallest incoming tuple for every destination. An edge becomes a proposal only when it is both choices. This reciprocal test prevents one source from being attached to a destination that prefers another predecessor, while naturally leaving unsupported tracklets unmatched.

Proposals are processed in ascending $(d_{ij}^{app}, g_{ij}, i, j)$ order with a disjoint-set data structure. Before joining components C_i and C_j , the method checks

$$\left(\bigcup_{u \in C_i} F_u \right) \cap \left(\bigcup_{v \in C_j} F_v \right) = \emptyset. \quad (4)$$

A proposal that violates Eq. (4) is rejected; otherwise the components are joined and inherit the earlier root identity. Finally, only track IDs are remapped. Frame indices, boxes, classes, confidence values, and the number of predictions are invariant. No interpolation fills temporal gaps.

This implementation does not use the candidate score as calibrated uncertainty, and it has no active re-observation action. With E eligible edges, exhaustive candidate construction costs $O(K^2)$, proposal sorting costs $O(E \log E)$, and union-find updates are near-linear in the number of accepted proposals. The explicit candidate graph requires $O(K + E)$ memory.

4. Experimental Protocol

4.1. Datasets, inputs, and frozen split

MMOT. We use all 50 sequences in the official MMOT test release [4]: 5,466 frames and 196,194 valid annotated boxes. Raw images are HWC `uint8` arrays containing eight spectral bands. Following the MMOT paper and the active official tracking loaders, pseudo-RGB is formed from one-based bands 5, 3, and 2, corresponding to zero-based indices $[4, 2, 1]$ in RGB order. The earlier local $[0, 1, 2]$ extraction was not equivalent; all appearance-dependent results in this paper were recomputed with the corrected channels.

Two input conditions are kept separate. The *oracle-AABB* condition converts each official oriented annotation to its axis-aligned envelope, assigns confidence 1.0, and gives every method the same boxes and classes. Ground-truth identity remains hidden until scoring. The *official detector* condition runs the MMOT authors’ YOLO11L-3ch checkpoint once at 1,280-pixel input, confidence 0.1, and NMS IoU 0.6. Its 193,977 detections over all 5,466 frames, including one frame with no detection, are cached and reused by every tracker and refiner.

Twelve MMOT sequences had appeared in our earlier analysis. Before examining the remaining results, we froze these as `legacy12` and assigned every other official test sequence to `confirmation38`; no sequence was selected by performance. The exact membership is listed in Appendix Appendix A. The rule and all thresholds remain unchanged across both partitions.

M3OT transfer.. M3OT [5] contributes four development streams (1-08T, 1-08, 2-08T, and 2-08) and four held-out streams (1-03T, 1-03, 2-03T, and 2-03). Suffix T denotes infrared. The temporal, geometry, appearance, and reciprocal settings were fixed from the development streams before held-out evaluation. Each stream is scored independently with the release’s vehicle filter and common oracle AABBs. We do not use the M3OT held-out result to retune the rule.

Both datasets are used for within-stream temporal identity. Neither protocol evaluates a global identity shared among different UAV cameras.

4.2. Upstream trackers and appearance model

ByteTrack, OC-SORT, and Deep OC-SORT are instantiated through BoxMOT commit `6edfa8ad` [16]. Each official fine-grained MMOT class is processed by an independent tracker instance, every released frame is passed in numeric order, and no area, height, or aspect-ratio filter is added by our adapter. Table 1 records the active settings. They are the historical BoxMOT AABB configurations used by this study, rather than the rotated-tracker settings in MMOT’s native benchmark code.

Table 1: Frozen upstream tracker settings. A dash denotes a parameter not used by that tracker. Deep OC-SORT additionally uses its default adaptive appearance weighting and camera-motion compensation.

Tracker	Det. gate	Match gate	Max age	Min hits	IoU gate	Δt
ByteTrack	0.45	0.80	30	3	0.30	–
OC-SORT	0.30	–	30	3	0.30	3
Deep OC-SORT	0.50	–	30	3	0.30	3

Tracklet descriptors use a fixed 128-dimensional ResNet-50 checkpoint from the OpenMMLab MOT17 BaseReID configuration. Its SHA-256 is reported in Appendix Appendix C. The test transform resizes crops to 128×256 and applies the checkpoint’s ImageNet normalization. The model is not trained or adapted on MMOT or M3OT. Deep OC-SORT receives per-detection embeddings from the same checkpoint; CoM3D-ACE then computes tracklet-level descriptors from the frozen output boxes.

4.3. Comparison methods

All refiners receive exactly the same saved upstream tracklets for a given tracker and input condition.

- **None** leaves every local identity unchanged.
- **Geometry greedy** takes every temporally eligible edge within 55 pixels, orders edges by endpoint distance, and applies the component frame-overlap constraint.
- **Partial Hungarian** uses candidates satisfying the same geometry and appearance gates. Its controlled cost is the mean of $d^{xy}/55$, $d^{app}/0.30$, and $g/30$, with a dummy unmatched cost of 1.0. Selected edges also pass the component constraint.
- **AFLink** uses the official StrongSORT implementation and checkpoint [9], with released settings $\text{thrT} = (0, 30)$, $\text{thrS} = 75$, and $\text{thrP} = 0.05$. Gaussian-smoothed interpolation is not applied.
- **CoM3D-ACE** is the reciprocal rule in Section 3.

The ablation additionally evaluates geometry-only reciprocal selection, geometry+ReID greedy linking, and reciprocal linking without the component guard. GIAOTracker is discussed as relevant global-linking work [8], but its authors’ public repository states that code release is not planned and does not provide the learned global-link model and weights needed for a faithful run. We therefore do not label a local approximation as official GIAOTracker.

4.4. Metrics and aggregation

TrackEval commit 12c8791b computes HOTA, its association component AssA, IDF1, CLEAR MOTA, and identity switches (IDSW). HOTA averages the standard localization-threshold curve; AssA measures association quality among matched detections. IDF1 is the F1 score after global one-to-one trajectory matching, and IDSW counts changes in the matched predicted identity over time. Higher is better for HOTA, AssA, IDF1, and MOTA; lower is better for IDSW.

Headline HOTA, AssA, and IDF1 values are equal-sequence means, so long sequences do not dominate. IDSW, FP, and FN are summed over exactly the same sequence set. We never recompute F1 from averaged precision and recall. The 38-sequence confirmation analysis uses paired sequence differences and 10,000 bootstrap resamples with seed 20260914. Repeated deterministic executions are reproducibility checks, not independent statistical samples.

For the link audit, an endpoint tracklet receives a post-hoc actor label only when its matched observations have a unique majority identity. Link coverage

uses the fixed set of source tracklets with at least one temporally eligible, non-overlapping successor within 30 frames. Link error is the false-link fraction among accepted edges whose two endpoint identities are auditable. Unknown edges remain in coverage but not in the error denominator. The sweep is descriptive and is not used to select a test threshold.

4.5. Integrity and runtime protocol

Before evaluation, exact GT-as-prediction and globally permuted-ID tests had to return HOTA, IDF1, and MOTA of 100 with IDSW 0. No-refinement had to reproduce the cached tracker output exactly. For every refiner, we check preservation of the (`frame`, `box`, `confidence`, `class`) multiset and reject aggregate reporting when one identity appears twice in a frame. Ground-truth actor identity is read only after predictions are frozen, for metrics and post-hoc edge labels.

Runtime is measured over the full MMOT test set on an NVIDIA RTX 5090 with PyTorch 2.9.1 and CUDA 12.8. BaseReID uses batch 128. Graph timing excludes disk I/O, upstream tracking, descriptor extraction, and metric evaluation; total post-tracking overhead adds descriptor and graph time and divides by 5,466 frames. These are offline replay measurements rather than flight-system latencies.

5. Results

5.1. Full-test refinement with oracle AABBs

Table 2 compares refiners on all 50 MMOT sequences with common oracle AABBs. Relative to no refinement, CoM3D-ACE raises IDF1 by 2.28, 2.29, and 2.18 percentage points for ByteTrack, OC-SORT, and Deep OC-SORT. The corresponding total IDSW reductions are 2,545, 1,678, and 585. Thus the fixed rule improves identity continuity for all three frozen upstream trackers, including the appearance-aware Deep OC-SORT.

The direct controls are stronger. Geometry-only greedy linking attains the best ByteTrack IDF1 (51.02), 1.30 points above CoM3D-ACE. Partial Hungarian attains the best OC-SORT and Deep OC-SORT IDF1 (46.19 and 84.00), 0.93 and 0.33 points above CoM3D-ACE. AFLink creates duplicate same-frame identities in 2, 3, and 4 sequences for ByteTrack, OC-SORT, and Deep OC-SORT, respectively; we leave its full-set rows unreported rather than repair them post hoc. The result supports improvement over no refinement, not accuracy superiority over global assignment.

Table 2: MMOT full-test temporal refinement with common oracle AABBs. HOTA, AssA, and IDF1 are equal-sequence means over 50 sequences; IDSW is the total. All refiners receive identical saved tracklets. Bold metric values mark the best valid refiner for each upstream tracker. AFLink is N/A when at least one sequence violates the no-duplicate-identity-per-frame invariant.

Upstream	Refiner	HOTA	AssA	IDF1	IDSW
ByteTrack	None	44.20	44.69	47.44	20,056
	Geometry greedy	45.60	46.97	51.02	15,484
	Partial Hungarian	45.55	46.90	50.78	16,462
	AFLink (official)	–	–	–	–
	CoM3D-ACE	45.19	46.34	49.72	17,511
OC-SORT	None	44.92	51.43	42.96	6,119
	Geometry greedy	45.90	53.68	45.41	4,122
	Partial Hungarian	46.26	54.45	46.19	3,547
	AFLink (official)	–	–	–	–
	CoM3D-ACE	45.92	53.64	45.25	4,441
Deep OC-SORT	None	81.32	82.06	81.49	1,962
	Geometry greedy	82.58	84.52	83.69	1,294
	Partial Hungarian	82.79	84.93	84.00	1,174
	AFLink (official)	–	–	–	–
	CoM3D-ACE	82.60	84.54	83.67	1,377

5.2. Full-test refinement with official detections

Table 3 repeats the comparison using detections from the official MMOT detector. CoM3D-ACE improves IDF1 by 1.65, 1.62, and 1.00 points and reduces IDSW by 1,389, 1,059, and 324. These gains are smaller than in the oracle-AABB condition but preserve the same direction under missed detections, false positives, and one zero-detection frame.

The strongest controlled rows remain geometry greedy for ByteTrack and partial Hungarian for OC-SORT and Deep OC-SORT. Official AFLink is valid over all 50 sequences only for Deep OC-SORT, where it changes IDF1 from 63.09 to 63.33; CoM3D-ACE reaches 64.08, while partial Hungarian reaches 64.17. The ByteTrack and OC-SORT AFLink aggregates remain N/A because three and one sequences, respectively, violate the output invariant.

5.3. Frozen confirmation partition

The 38 sequences not included in our earlier MMOT analysis provide a frozen confirmation partition. Table 4 reports CoM3D-ACE minus no-

Table 3: MMOT full-test temporal refinement from one shared official YOLO11L-3ch detection cache. Aggregation and invariant handling match Table 2.

Upstream	Refiner	HOTA	AssA	IDF1	IDSW
ByteTrack	None	34.65	42.29	38.32	10,324
	Geometry greedy	35.64	44.14	40.76	7,908
	Partial Hungarian	35.58	44.11	40.56	8,405
	AFLink (official)	–	–	–	–
	CoM3D-ACE	35.40	43.75	39.97	8,935
OC-SORT	None	35.89	47.62	36.46	4,263
	Geometry greedy	36.51	49.40	38.14	2,959
	Partial Hungarian	36.77	49.97	38.74	2,668
	AFLink (official)	–	–	–	–
	CoM3D-ACE	36.53	49.24	38.09	3,204
Deep OC-SORT	None	53.92	67.00	63.09	1,370
	Geometry greedy	54.25	67.71	64.01	1,008
	Partial Hungarian	54.36	67.95	64.17	957
	AFLink (official)	54.00	67.16	63.33	1,342
	CoM3D-ACE	54.31	67.84	64.08	1,046

refinement deltas. All six paired IDF1 intervals remain above zero. Under oracle AABBs, every ByteTrack sequence improves and only one OC-SORT sequence worsens. Under detector inputs, improvement counts remain 33, 36, and 27 of 38 for the three trackers. This reduces the likelihood that the positive no-refinement comparison is confined to the original 12 sequences, while not changing the stronger-baseline conclusion above.

Table 4: CoM3D-ACE minus no-refinement paired changes on the frozen confirmation38 partition. Confidence intervals are 95% sequence-bootstrap intervals with 10,000 resamples and seed 20260914. The final column gives improved/tied/worsened IDF1 sequence counts.

Input	Tracker	Δ HOTA	Δ AssA	Δ IDF1 (95% CI)	Δ IDSW	I/T/W
Oracle	ByteTrack	+1.05	+1.71	+2.38 [1.85, 2.99]	-2,147	38/0/0
	OC-SORT	+0.91	+2.07	+2.15 [1.54, 2.84]	-1,185	37/0/1
	Deep OC-SORT	+1.28	+2.49	+2.29 [1.37, 3.57]	-466	31/4/3
Official det.	ByteTrack	+0.79	+1.48	+1.71 [1.20, 2.26]	-1,142	33/1/4
	OC-SORT	+0.58	+1.46	+1.51 [1.07, 2.04]	-759	36/0/2
	Deep OC-SORT	+0.44	+0.92	+1.10 [0.48, 2.00]	-268	27/5/6

5.4. Component ablation and accepted-link risk

Table 5 isolates the rule components under the oracle-AABB protocol. Geometry-only reciprocal selection accounts for most of the gain over no refinement. Adding the fixed appearance gate yields a small additional increase. The no-guard and full CoM3D-ACE rows are identical for every reported metric: no proposed MMOT merge activates the component-overlap guard. The guard remains an output-validity rule, but this dataset does not demonstrate a metric contribution from it. Geometry+ReID greedy linking is stronger than CoM3D-ACE for all three trackers.

Table 5: Temporal component ablation on all 50 MMOT sequences with oracle AABBs. IDF1 is the equal-sequence mean and IDSW is the total. All rows reuse the same upstream tracklets and candidate universe.

Variant	ByteTrack		OC-SORT		Deep OC-SORT	
	IDF1	IDSW	IDF1	IDSW	IDF1	IDSW
None	47.44	20,056	42.96	6,119	81.49	1,962
Geometry reciprocal	49.62	17,581	45.08	4,551	83.63	1,392
+ReID reciprocal, no guard	49.72	17,511	45.25	4,441	83.67	1,377
CoM3D-ACE	49.72	17,511	45.25	4,441	83.67	1,377
Geometry+ReID greedy	51.80	15,064	46.25	3,481	83.99	1,149
Partial Hungarian	50.78	16,462	46.19	3,547	84.00	1,174

Figure 2 shows the descriptive link error-coverage trajectories. At the fixed 0.30 gate, CoM3D-ACE covers 36.6%, 37.9%, and 26.1% of eligible source tracklets for ByteTrack, OC-SORT, and Deep OC-SORT; its auditable-link error rates are 44.6%, 35.1%, and 39.2%. The corresponding partial-Hungarian values are 54.0%/52.9%, 49.5%/25.6%, and 32.7%/35.3%. For ByteTrack, 3,909 accepted CoM3D-ACE links have an unknown endpoint label and are excluded only from the error denominator. The high and tracker-dependent error rates preclude a calibrated risk claim. Tighter points on the curves are not proposed post-test operating points.

5.5. Held-out M3OT transfer

The same fixed rule has the opposite outcome on held-out M3OT (Table 6). It improves the four development streams but decreases equal-sequence-mean IDF1 by 4.00 points for ByteTrack and 0.55 for OC-SORT on the four held-out streams. All ten admitted held-out links with known identity are false,

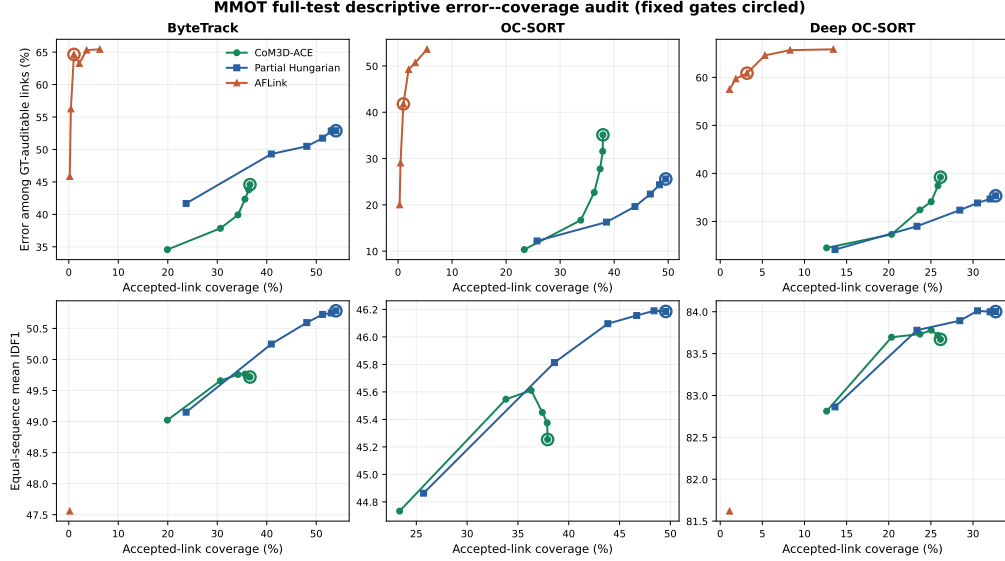


Figure 2: Full-test accepted-link error versus coverage for CoM3D-ACE, partial Hungarian, and official AFLink. Coverage uses a fixed eligible-source denominator; error uses only post-hoc auditable accepted links. Markers identify the predeclared 0.30 CoM3D-ACE/Hungarian and 0.05 AFLink gates. Curves are descriptive and were not used for test-time threshold selection. The plot was generated directly from frozen CSVs by a reproducible script prepared with OpenAI Codex assistance; no value was generated or altered by an AI model.

while only three and five correct fragment opportunities exist for ByteTrack and OC-SORT. Nine false links occur on IR streams and one on RGB. This association is consistent with a modality and domain shift, but does not establish either as the sole cause. We retain the failure and do not add a dataset-specific bypass.

5.6. Object scale, runtime, and qualitative audit

Table 7 verifies the small-object composition. Across all 50 sequences, 53.4% of boxes have area below 32^2 pixels, 45.9% lie from 32^2 to 96^2 , and the median short side is 21 pixels. The confirmation38 partition is not easier by this proxy: it has a larger sub- 32^2 fraction and a smaller median short side than legacy12.

Post-tracking runtime is dominated by appearance extraction (Table 8). BaseReID plus graph processing costs 46.9 ms/frame for ByteTrack and approximately 24.5 ms/frame for OC-SORT and Deep OC-SORT. Graph construction and decisions alone total 7.99, 2.09, and 1.21 seconds, respectively,

Table 6: Fixed-policy M3OT development and held-out transfer. Values are equal-sequence means over four streams per split; IDSW and post-hoc link counts are totals. “Opportunities” counts auditable same-actor fragment pairs in the fixed candidate universe.

Split	Tracker	HOTA base→ref.	IDF1 base→ref.	IDSW base→ref.
Development	ByteTrack	93.28→94.16	96.89→98.49	20→10
	OC-SORT	96.31→97.55	96.10→98.26	21→9
Held-out	ByteTrack	88.31→85.55	96.77→92.77	6→6
	OC-SORT	94.05→93.60	95.39→94.84	5→5

Split	Tracker	Correct links	False links	Opportunities
Development	ByteTrack	10	3	15
	OC-SORT	12	5	19
Held-out	ByteTrack	0	7	3
	OC-SORT	0	3	5

Table 7: Native MMOT annotation-box distribution. Area bins are fixed COCO boundaries; percentages use all valid boxes in each partition.

Partition	Boxes	$A < 32^2$	$32^2 \leq A < 96^2$	Side < 16	Median side
All 50	196,194	53.4%	45.9%	43.1%	21 px
Legacy 12	30,666	47.9%	51.7%	38.5%	23 px
Confirm. 38	165,528	54.4%	44.9%	44.0%	19 px

over the 5,466-frame release. The shared detector separately requires 226.15 seconds, or 41.4 ms/frame.

Table 8: Full-test post-tracking runtime. ReID uses an RTX 5090 and batch 128; graph time is in-memory CPU candidate construction and decision time. The final column combines ReID and graph time over 5,466 frames.

Tracker	Crops	Desc.	ReID (s)	Cand.	Graph (s)	Total (ms/fr.)
ByteTrack	29,281	13,836	248.3	864,743	7.99	46.9
OC-SORT	15,819	8,150	131.6	216,508	2.09	24.5
Deep OC-SORT	15,490	5,663	132.9	76,253	1.21	24.5

Figure 3 complements aggregate scores with one correct and one false accepted edge selected by a deterministic rule from confirmation38. The correct example links ByteTrack bus IDs 1 and 3 across frames 56–58 in data41-1; both endpoints map post hoc to actor 18 at purity 1.0. The failure links Deep OC-SORT car IDs 40 and 48 across frames 393–397 in data24-1,

but their post-hoc actors differ. The pair illustrates why a plausible short-gap, close-appearance edge does not by itself guarantee identity.

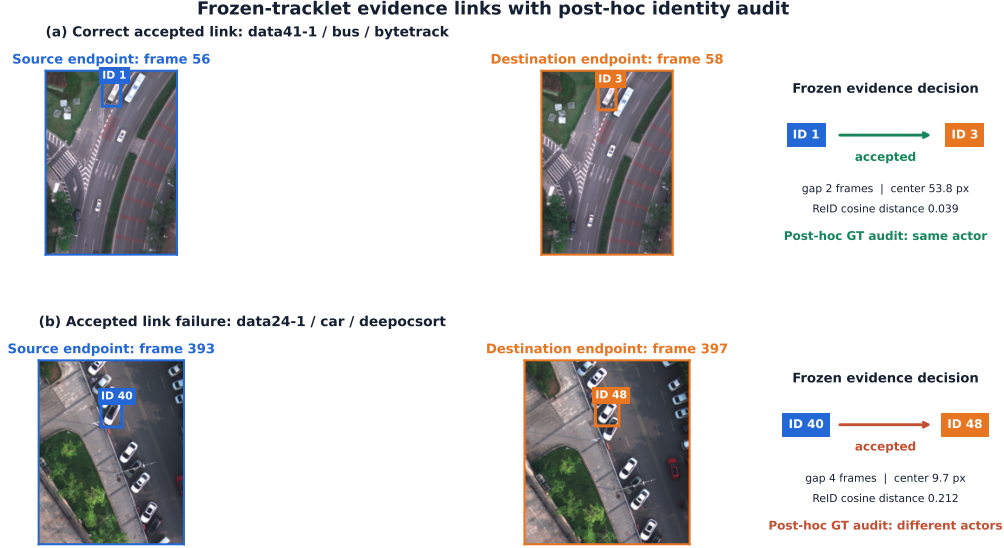


Figure 3: Actual CoM3D-ACE accepted-link examples from the frozen MMOT confirmation38 predictions. Top: a correct ByteTrack bus link in data41-1. Bottom: a false Deep OC-SORT car link in data24-1. Actor labels are added only after linking for audit. Original pseudo-RGB frames are unchanged except for deterministic crops, boxes, and labels. The rendering script was prepared with OpenAI Codex assistance and its selected rows are recorded in a checksummed manifest.

6. Discussion

The full-test results support one bounded conclusion: a fixed reciprocal post-linker improves the unchanged output of three tracker families on MMOT. The positive direction survives both oracle and official-detector inputs and the separately frozen confirmation38 partition. Because every compared method receives identical detections and local tracklets, the measured differences are attributable to identity refinement rather than detector retraining, tracker tuning, interpolation, or sequence filtering.

The strong controls materially qualify that conclusion. CoM3D-ACE does not lead the full-test tables. A geometry-only greedy rule gives ByteTrack greater IDF1, and partial Hungarian assignment gives OC-SORT and Deep OC-SORT greater IDF1. The geometry+ReID greedy ablation is also stronger.

Reciprocity therefore acts as a deterministic selection rule with explicit unmatched output, not as evidence that local mutual choice is globally optimal. Likewise, the component-overlap guard is logically useful for maintaining valid MOT output, but its identical ablation row shows that it is inactive on these MMOT proposals.

The link audit explains why trajectory gains and edge-level error need not rank methods identically. One accepted edge can join long fragments and improve IDF1 substantially, while another creates a costly false merge. Endpoint majority labels are also unavailable for many ByteTrack edges, so the auditable-error denominator is narrower than coverage. We therefore retain trajectory metrics beside the error-coverage plot and avoid calling the fixed gate calibrated uncertainty or guaranteed abstention.

The M3OT failure is the clearest transfer boundary. Its held-out streams offer few correct fragment opportunities, yet the fixed MMOT-beneficial rule accepts ten false links. Fixed pixel displacement and a MOT17 appearance descriptor do not normalize camera scale, modality, motion, or crop quality. A future method could learn or calibrate these factors on representative development data, but doing so after observing the current tests would define a new method requiring new untouched evaluation. No such adaptation is introduced here.

7. Limitations

First, this study evaluates temporal refinement within individual UAV streams. It does not assign a persistent identity across cameras, reconstruct 3D geometry, coordinate multiple UAVs, or command follow-up observations. The method name is retained for continuity with the implementation, but those capabilities are outside the claim tested here.

Second, the candidate rule is heuristic and domain dependent. Its 55-pixel geometry gate is not scale normalized, its BaseReID descriptor was trained on MOT17 rather than aerial multispectral imagery, and its ranking prioritizes endpoint geometry. The adverse M3OT result shows that a fixed gate validated in one setting cannot be presumed to transfer.

Third, 12 of the 50 MMOT test sequences were present in our earlier analysis. We expose them as legacy12 and separately report the remaining confirmation38, but the complete test set is not a wholly unseen benchmark. The confirmation intervals quantify stability against sequence sampling, not performance under all possible aerial domains.

Fourth, the official AFLink integration produces invalid duplicate identities on several sequences under our no-interpolation output contract, so most full-set AFLink aggregates are N/A. A faithful official GIAOTracker global-link run is blocked by unavailable implementation and compatible weights. These missing rows limit comparison with learned global linkers; the controlled greedy and partial-Hungarian rows remain the strongest executable baselines in this protocol.

Finally, detector-input experiments use one checkpoint supplied by the MMOT authors. Runtime is measured as offline replay on one workstation. Neither result establishes real-time onboard operation, energy cost, communication cost, or robustness to another detector family.

8. Conclusion

CoM3D-ACE is evaluated here as a deterministic evidence-graph refiner for frozen aerial tracklets. Across all 50 MMOT test sequences, it improves HOTA, AssA, IDF1, and identity switches over no refinement for ByteTrack, OC-SORT, and Deep OC-SORT under both oracle and official-detector inputs. The positive IDF1 change also persists on the 38 sequences outside our earlier analysis. At the same time, direct greedy and partial-Hungarian controls usually score higher, the component guard is inactive on MMOT, and held-out M3OT transfer is negative. The resulting contribution is a reproducible post-tracking improvement with explicit scope and failure evidence, not a universal or globally optimal association method.

CRedit authorship contribution statement

Jong-Chan Park: Conceptualization, Methodology, Investigation, Writing – original draft. So-Hee Lim: Visualization, Writing – review & editing. Sujin Park: Visualization, Writing – review & editing. Sang-Min Choi: Formal analysis, Writing – review & editing. Gun-Woo Kim: Formal analysis, Writing – review & editing.

Data and code availability

The MMOT [4] and M3OT [5] datasets are publicly available from their official sources: the MMOT repository, the MMOT dataset page, and the M3OT repository. The implementation, frozen split manifest, configurations,

and reproduction instructions are planned for public release through a GitHub repository. No repository URL is claimed before that release is available.

Declaration of generative AI and AI-assisted technologies in the writing process

During preparation of this work, the authors used OpenAI Codex to assist with manuscript organization and language revision, implementation audits, and reproducible analysis and visualization scripts. All scripts were executed on frozen inputs; numerical outputs and figure source rows were checksummed and reviewed by the authors. No generative model created or altered evaluation labels, predictions, or primary dataset imagery. The authors reviewed and edited the resulting material and take full responsibility for the content of the publication.

Appendix A. MMOT reporting partitions

Table A.9 records the frozen split used only for reporting. `Confirmation38` contains every official test sequence not in the previously analyzed `legacy12` set.

Table A.9: Frozen MMOT test partition. Frame counts and source hashes are stored in the accompanying split manifest. No sequence was assigned by performance.

Partition	Sequence identifiers
Legacy12	data28-1, data28-2, data28-3, data28-4, data28-5, data28-6, data30-2, data30-3, data30-4, data30-5, data30-9, data30-10
Confirm. 38	data23-1, data24-1, data27-1, data31-1, data33-1, data34-1, data34-2, data34-3, data36-4, data36-5, data36-10, data36-11, data36-12, data36-13, data37-1, data37-2, data37-10, data37-11, data37-12, data39-1, data39-2, data39-3, data39-6, data40-3, data40-4, data40-5, data41-1, data42-1, data42-2, data42-3, data46-11, data46-12, data47-1, data47-2, data47-3, data47-4, data48-1, data49-2

Appendix B. Fixed operating-point link audit

Table B.10 gives the fixed points highlighted in Fig. 2. AFLink trajectory metrics are N/A when any sequence produces invalid duplicate same-frame identities. Its edge-level audit remains descriptive and does not repair those trajectories.

Table B.10: Full-test fixed operating-point link audit with oracle AABBs. Coverage and auditable-link error use the denominators defined in Section 4. IDF1 is the corresponding valid trajectory score.

Tracker	Refiner	Gate	Coverage	Link error	IDF1
ByteTrack	CoM3D-ACE	0.30	36.6%	44.6%	49.72
	Partial Hungarian	0.30	54.0%	52.9%	50.78
	AFLink (official)	0.05	1.0%	64.6%	–
OC-SORT	CoM3D-ACE	0.30	37.9%	35.1%	45.25
	Partial Hungarian	0.30	49.5%	25.6%	46.19
	AFLink (official)	0.05	1.0%	41.8%	–
Deep OC-SORT	CoM3D-ACE	0.30	26.1%	39.2%	83.67
	Partial Hungarian	0.30	32.7%	35.3%	84.00
	AFLink (official)	0.05	3.2%	60.9%	–

Appendix C. Reproducibility record

The release manifest records full revisions and checksums. Their compact prefixes are:

- MMOT test revision: 57dfe1521ee2;
- BaseReID SHA-256: 4bf6b63d4923;
- MMOT YOLO11L-3ch SHA-256: 6d9f66ed17eb;
- AFLink SHA-256: b35cbecdd3ac;
- TrackEval commit: 12c8791b303e;
- BoxMOT commit: 6edfa8ad7dc2.

Evaluator sanity tests return HOTA/IDF1/MOTA 100 and IDSW 0 for both exact GT predictions and a global one-to-one permutation of prediction identities. Legacy12 rows in the standalone and full-50 executions match at absolute tolerance zero. The source artifact package contains commands, source and model hashes, successful manifests, failed-run records, per-sequence CSVs, and a SHA-256 index.

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